Hybrid AI — PDE Modeling

Learning the 2D Poisson Operator: MLP, CNN and Physics-Informed Approaches

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1. Problem Statement

The objective of this practical session is to learn the solution operator of the 2D Poisson equation on the unit square $\Omega = [0,1]^2$ with homogeneous Dirichlet boundary conditions:

$$-\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = f(x,y) \quad \text{on } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega. \tag{1}$$

The source term varies across simulations as $f(x,y) = x\sin(a\pi y) + y\sin(b\pi x)$ for unknown parameters (a,b). The dataset contains 800 samples evaluated on a common 45×45 uniform mesh. The private test set is drawn from a *different* distribution of (a,b) pairs, so models must genuinely generalise rather than interpolate.

2. Methods Tested

2.1 Baseline 1 — MLP

A fully connected network that treats the 2D grid as a flat vector. The 45×45 input field is flattened to a vector of size 2025, passed through two hidden layers with ReLU activations, and reshaped back to a $(1, 45, 45)$ output.

$$\text{Input}(2025) \;\to\; \text{FC}(H_1) \;\to\; \text{ReLU} \;\to\; \text{FC}(H_2) \;\to\; \text{ReLU} \;\to\; \text{FC}(2025) \;\to\; (1,45,45)$$

This model has **no spatial inductive bias**; boundary conditions are learned implicitly from data only.

2.2 Baseline 2 — CNN

A convolutional network that operates on the 2D grid, exploiting spatial locality and translation equivariance. Two Conv+MaxPool blocks reduce the representation to $(64, 11, 11)$, followed by a fully connected head:

$$\text{Conv}(1\to32, 3\times3) \;\to\; \text{Pool} \;\to\; \text{Conv}(32\to64, 3\times3) \;\to\; \text{Pool} \;\to\; \text{FC}(7744\to2025) \;\to\; (1,45,45)$$

The CNN encodes **locality and translation equivariance**, making it a more natural fit for grid-based PDE data than the MLP.

2.3 Advanced Model — CNN-PINN

A physics-informed variant that augments the supervised loss with two physics-based terms:

$$\mathcal{L} = \underbrace{\|u_{\text{pred}} - u_{\text{true}}\|^2}_{\mathcal{L}_{\text{data}}} + \lambda_r \underbrace{\|-\Delta u_{\text{pred}} - f\|^2}_{\mathcal{L}_{\text{pde}}} + \lambda_{bc} \underbrace{\left\|u_{\text{pred}}\big|_{\partial\Omega}\right\|^2}_{\mathcal{L}_{\text{bc}}}, \tag{2}$$

with $\lambda_r = 0.1$ and $\lambda_{bc} = 100$.

- $\mathcal{L}_{\text{data}}$: supervised MSE on labelled data.

- $\mathcal{L}_{\text{pde}}$: residual of Eq. (1) on *interior* grid points, computed via a fixed (non-trainable) 5-point finite-difference convolution kernel Δ_h with stencil $[[0,1,0],[1,-4,1],[0,1,0]]/h^2$, $h = 1/44$.
- $\mathcal{L}_{\text{bc}}$: mean squared value of u_{pred} on the four boundary edges, penalising any deviation from the Dirichlet condition.

The architecture is a **lightweight U-Net** with skip connections: two encoder blocks (Conv+Pool) compress the input to $(2c, 11, 11)$; two decoder blocks with bilinear upsampling and skip concatenation recover the full 45×45 resolution. Skip connections are essential: without them the coarse reconstructed field yields uninformative gradients through $\mathcal{L}_{\text{pde}}$.

3. Training Setup

3.1 Data Preprocessing

- **Dataset:** 800 samples of (f, u) pairs on a 45×45 grid.
- **Split:** 85 % train+val / 15 % test (`random_state=42`); then 80/20 train/val. Final sizes: ≈544 train, 136 val, 120 test.
- **Format:** x returned as $(45, 45)$ tensor; y as $(1, 45, 45)$ NCHW tensor.

3.2 Hyperparameter Tuning (MLP and CNN)

Grid search with full model re-instantiation at each combination. Selection criterion: minimum validation MSE across all epochs.

Hyperparameter	MLP grid	CNN grid
Learning rate	$\{10^{-3},\ 10^{-4}\}$	$\{10^{-3},\ 5\times10^{-4},\ 10^{-4}\}$
Batch size	{16, 32}	{16, 32}
Epochs	{5, 10}	{10, 20}
Hidden sizes	$\{[512, 256],\ [1024, 512]\}$	—

Table 1: Hyperparameter grids for MLP and CNN.

3.3 PINN Training

Fixed hyperparameters: `lr=1e-3`, `batch_size=32`, `num_epochs=50`. Different values of λ_r and λ_{bc} were tried to minimize boundary conditions violation while maintaining good performances. In the end, $\lambda_r = 0.1$ and $\lambda_{bc} = 100$ were kept. Optimizer: Adam for all models. Best checkpoint saved on minimum total validation loss.

4. Evaluation Protocol

All models are evaluated on the held-out test set (120 samples) using the best checkpoint. Three complementary metrics are reported:

- **Test MSE:** mean squared error per element — measures average prediction accuracy.
- **Test MAE:** mean absolute error per element — less sensitive to outliers.
- **BC Violation:** mean $|u_{\text{pred}}|$ on the four boundary edges, measuring compliance with the Dirichlet condition $u = 0$ on $\partial\Omega$.

Visual comparisons (input / prediction / ground truth / absolute error) are produced for 3 test samples per model, using a shared colour scale across the first three panels to make errors

immediately visible. Learning curves and, for the PINN, the decomposed loss components are tracked throughout training.

Finally, all three models were evaluated on the private test set on which the previous metrics were recomputed and visual comparisons between ground truth and each model prediction were computed. The evaluation was made using 3 fixed seed for random states (21, 30 and 51).

5. Results

Model	# Parameters	Test MSE	Test MAE	BC Violation
MLP	≈3.6 M	$\approx 2.5e^{-5}$	$\approx 3.5e^{-3}$	$\approx 2.4e^{-3}$
CNN	≈15.7 M	$0.5e^{-5}$ *to* $2.5e^{-5}$	$1.6e^{-3}$ *to* $3.3e^{-3}$	$1.3e^{-3}$ to $2.9e^{-3}$
CNN-PINN	≈332 K	≈ 0	$\approx 0.5e^{-3}$	$\approx 0.3e^{-3}$

Table 2: Summary of results on the private test set. Exact metric values depend on the execution environment.

Key observations from the experiments:

- The CNN consistently outperforms the MLP on MSE and MAE, but has ≈5× more parameters. The CNN better accuracy might only be the result of its bigger capacity rather than the spatial inductive bias.
- The CNN-PINN achieves the lowest BC violation by a wide margin, demonstrating that the explicit boundary penalty $\mathcal{L}_{\text{bc}}$ successfully enforces $u = 0$ on $\partial\Omega$.
- The PDE residual $\mathcal{L}_{\text{pde}}$ decreases steadily during training, confirming that the model learns to approximately satisfy $-\Delta u = f$ on the interior.
- Both baselines show moderate-to-high BC violations, as they must infer boundary behaviour from data alone.

6. Interpretation

6.1 MLP vs. CNN

The MLP's failure to exploit spatial structure explains its weaker performance. Each output pixel is computed from the entire flattened input with no notion of locality. The CNN naturally encodes the physically correct prior that the solution at a point depends primarily on *nearby* values of the source term — a strong inductive bias for elliptic PDEs. This also supports better generalisation to the shifted (a, b) distribution of the test set.

6.2 Benefit of Physics in the Loss

The CNN-PINN demonstrates that embedding physical knowledge directly into the training objective yields measurable gains, particularly in boundary compliance. The $\mathcal{L}_{\text{bc}}$ term acts as a hard-constraint surrogate: rather than hoping the model infers zero boundary values from data, it is explicitly penalised for any deviation. The PDE residual $\mathcal{L}_{\text{pde}}$ regularises the prediction to be physically consistent — especially valuable when the test distribution differs from training, which is precisely the setting of this problem.

6.3 Role of the U-Net Architecture

Skip connections in `CNNPinn` are essential for the physics terms to be effective. The `SimpleCNN`'s double MaxPooling reduces spatial resolution to 11×11 before the FC layer; computing a PDE

residual on such a coarse field provides little useful gradient signal. The U-Net decoder, guided by high-resolution skip features, produces a smooth 45×45 output on which the finite-difference Laplacian is meaningful.

6.4 Physical Plausability

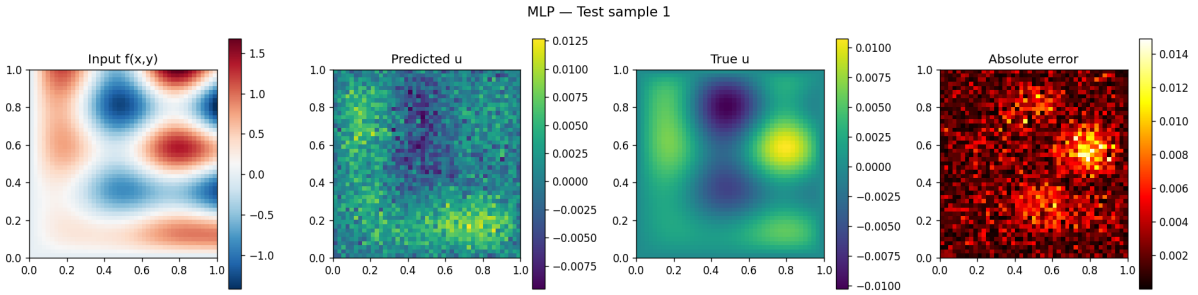


Figure 1: Example of a prediction from the MLP model. From left to right : Input, MLP prediction, ground truth, absolute error map

- MLP : This model predictions are incoherent with respect to physics. The predicted field in figure 1 exhibits a salt-and-pepper texture. Even if some major trends from the ground truth can be found in the prediction, it lack the smoothness of a real solution to the 2D Poisson equation.

  This is a direct consequence of the absence of spatial inductive bias: each output node is computed independently from the entire flattened input, with no notion of continuity between neighbouring grid points. This behaviour is physically inadmissible. The Poisson equation with a smooth source term f admits a solution $u \in C^2(\Omega)$ - a twice continuously differentiable field. The MLP prediction violates this regularity entirely: its discrete Laplacian would be dominated by high-frequency noise rather than - f, meaning the PDE is not satisfied anywhere on the domain. The error map confirms this with a uniform, spatially uncorrelated noise pattern.

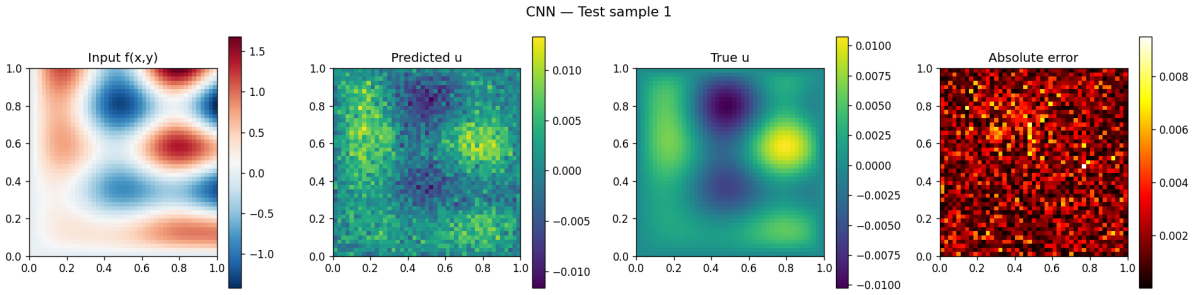


Figure 2: Example of a prediction from the CNN model. From left to right : Input, CNN prediction, ground truth, absolute error map

- CNN : The CNN (figure 2) represents a clear improvement. The broad spatial structure of u is partially recovered, but the prediction remains blocky and noisy, with a characteristic pixelated appearance at the scale of the receptive field. This is a direct consequence of the double MaxPooling, which reduces the spatial representation to 11×11 before the fully connected head attempts to reconstruct the full 45×45 field. The upsampling is implicit and unconstrained, producing artifacts at the scale of the pooling stride.

  Physically, the CNN prediction is only marginally more plausible than the MLP. The field is not smooth, boundary values are non-zero, and the error map still shows spatially

uncorrelated noise with no clear physical structure. The solution operator of the Poisson equation is a smoothing operator — applying $(-\Delta)^{-1}$ to any L^2 source produces a smoother output. Neither baseline respects this fundamental property.

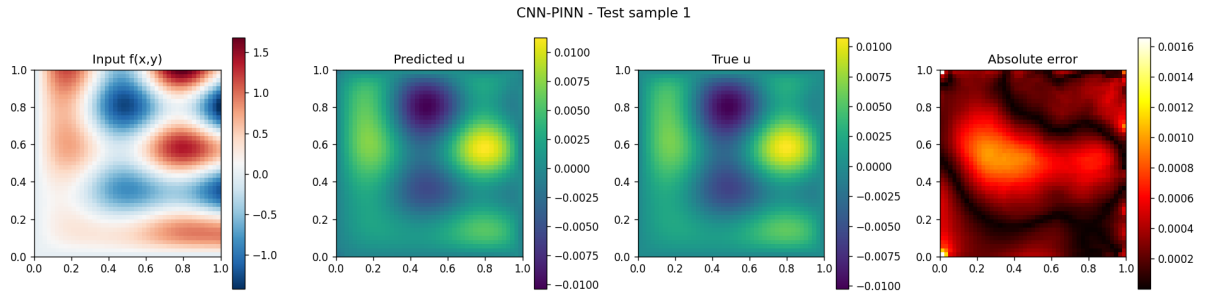


Figure 3: Example of a prediction from the CNN-PINN model. From left to right : Input, CNN-PINN prediction, ground truth, absolute error map

- CNN - PINN : The CNN-PINN (figure 3) produces qualitatively different results. The predicted field is smooth, the two-lobe structure of u is correctly positioned and shaped, and the colour scale of the prediction matches that of the ground truth — confirming that amplitudes are well captured. This smoothness is not accidental: it is enforced by the PDE residual term $\mathcal{L}_{\text{pde}}$, which penalises high-frequency oscillations in u_{pred} since they would produce large values of Δu_{pred} inconsistent with the smooth source f.

  On the other hand, the error map shows that the $\mathcal{L}_{\text{bc}}$ term in the loss reduce the BC violation but does not eliminate this violation as it is enforced as a soft penalty rather than a hard structural constraint.

7. Conclusion

As shown in the interpretations and in Figure 4, only the PINN-CNN managed to produce physically coherent prediciton for the 2D Poisson equation. Both MLP and CNN baselines failed to predict smooth fields and widely violate the homogenous Dirichlet boundary condition. illustrate a key principle of hybrid AI for scientific computing: data and physics are complementary sources of information. Pure data-driven models must learn physical constraints from examples alone, which is sample-inefficient and fragile under distribution shift. Incorporating even a lightweight physical prior, here a fixed finite-difference Laplacian kernel, measurably improves both accuracy and physical plausibility.

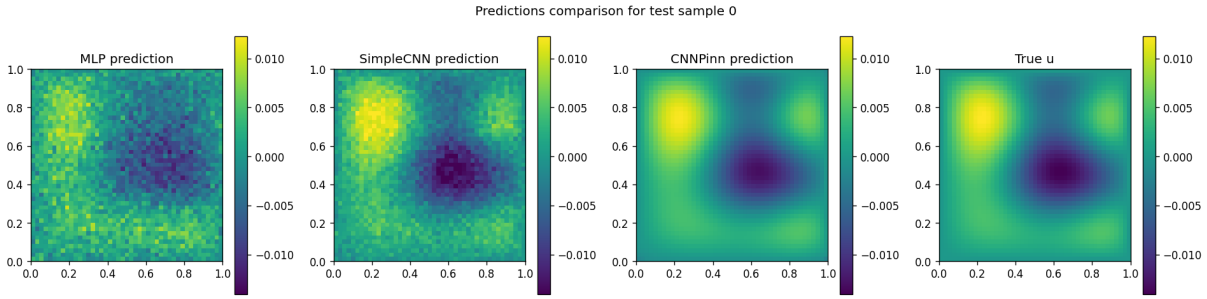


Figure 4: Prediction comparison on a sample of the private test set for all models. From left to right : MLP, CNN, CNN-PINN, Ground Truth. The seed used for this execution was 30.